

The end of software engineering?

How AI Agents Are Changing the Way Software Is Designed, Engineered & Delivered

Audience

MSc CS & Mathematics Students | CEG, Anna University | Industry Connect Seminar

Overview

For decades, software engineering has followed a familiar path: **Requirements >> Design >> Code >> Test >> Deploy**.

But now **AI agents can increasingly write code, use tools, run tests, debug failures, navigate codebases and work autonomously for hours**. So, if AI can write the code, then what exactly should a software engineer learn and do?

This session aims to explore how the engineering abstraction is moving **from writing software to engineering the systems that produce software**.

The session includes

- **Interactive Discussions:** Polls, questions, thought experiments and challenges throughout the session.
- **Career Perspective:** A practical view of which skills are becoming less valuable, which are becoming more valuable, and where students can build an advantage.
- **Live AI & Coding-Agent Demonstrations**
See AI agents plan, code, test, debug and iterate on real software tasks.

Topics

1. From Coding to Agentic Engineering

How we moved from **LLMs >> Copilots >> Coding Agents >> Agentic Systems**, and why this is more than just another developer tool.

2. The New Software Engineering Stack

Understand the emerging architecture: **Model >> Agent >> Loop >> Graph >> Harness >> Reliable Software**

What are agents actually doing, and what makes them capable of working autonomously?

3. The Engineer's New Job

When code becomes cheaper to produce, engineering value moves to: **Specification, Architecture, Context, Orchestration, Verification, Judgment**

Why **specification engineering/documentation** may matter more than prompt engineering.

4. Can We Trust AI-Generated Software?

AI-generated software is probabilistic and can fail in unexpected ways.

How do we build systems that can **test, evaluate, observe, verify and recover** from AI failures?

5. The Software Engineer of Tomorrow

What is the state of today's job market? What skills should students build today to remain valuable in an AI-native software industry? Why CS/Maths/SWE fundamentals still matter, perhaps more than ever; **Algorithms, Probability, Statistics, Optimization, Databases, Distributed Systems, System Design, User Experience Design, Security, Formal verification**

AI doesn't make these concepts obsolete. They become the vocabulary for engineering AI systems.

Speaker

Vikram Singh

Founder & Product Architect, Zestifai Technologies

Technology entrepreneur and product architect focused on building ai-powered business systems. He has over two decades of experience in enterprise software engineering, product development, digital transformation, and artificial intelligence.

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